

# palm-scan Rev 2 – palm-vein door lock

ESP32-S3 + DFRobot AI10 palm sensor + 12 V solenoid. Net names are global labels:  
the same name on any page is the same wire. See NETLIST.md for the table.

## POWER INPUT & POWER LED



File: power\_in.kicad\_sch

## RESET, BOOT & DEBUG UART



File: reset\_boot.kicad\_sch

## DISPLAY & STATUS LEDS



File: display.kicad\_sch

## 3.3V & 5V REGULATORS



File: power\_reg.kicad\_sch

## USB-C



File: usb.kicad\_sch

## EXIT BUTTON



File: exit.kicad\_sch

## ESP32-S3 MCU



File: mcu.kicad\_sch

## LOCK DRIVER & SENSOR



File: lock.kicad\_sch

## BUZZER & POWER FLAGS



File: buzzer.kicad\_sch

Palm-vein door lock – ESP32-S3, MOSFET lock driver, 12 V adapter

Sheet: /  
File: palm-scan.kicad\_sch

**Title: palm-scan Rev 2 – ROOT**

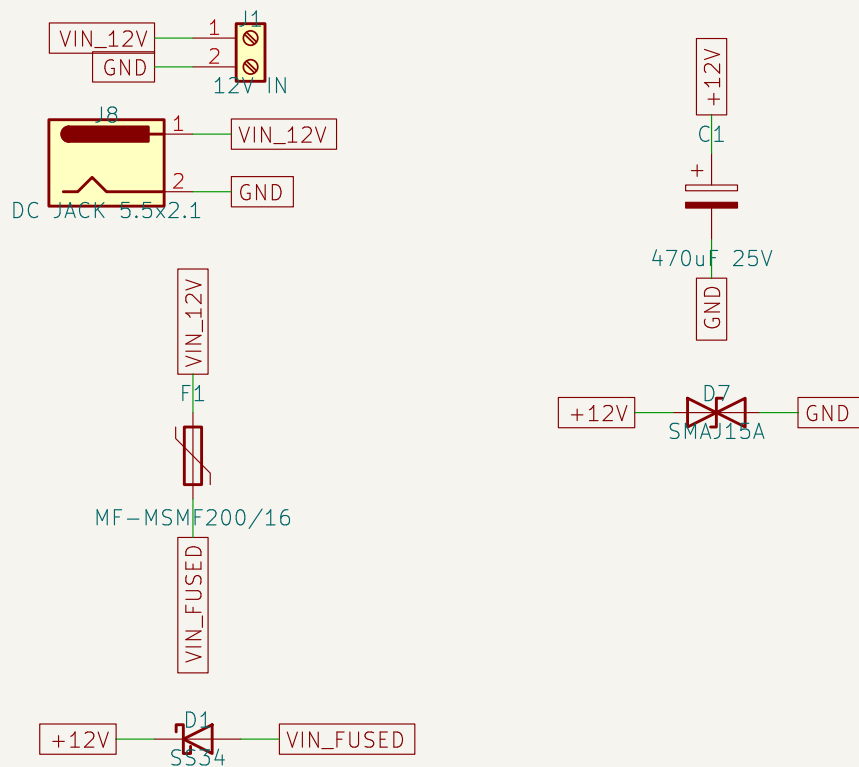
Size: A4      Date: 2026-09-13

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**Rev: 2**

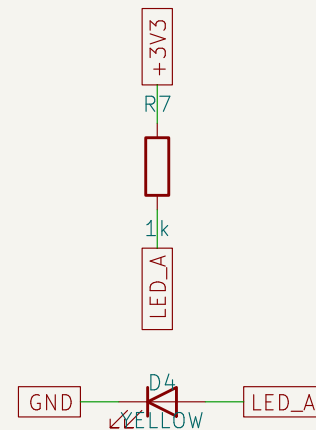
Id: 1/10

## POWER INPUT



Use J1 OR J8 – never both. Centre pin of J8 is +.

## POWER LED



Palm-vein door lock – ESP32-S3, MOSFET lock driver, 12 V adapter

Sheet: /POWER INPUT & POWER LED/

File: power\_in.kicad\_sch

**Title: palm-scan Rev 2 – POWER INPUT & POWER LED**

Size: A4

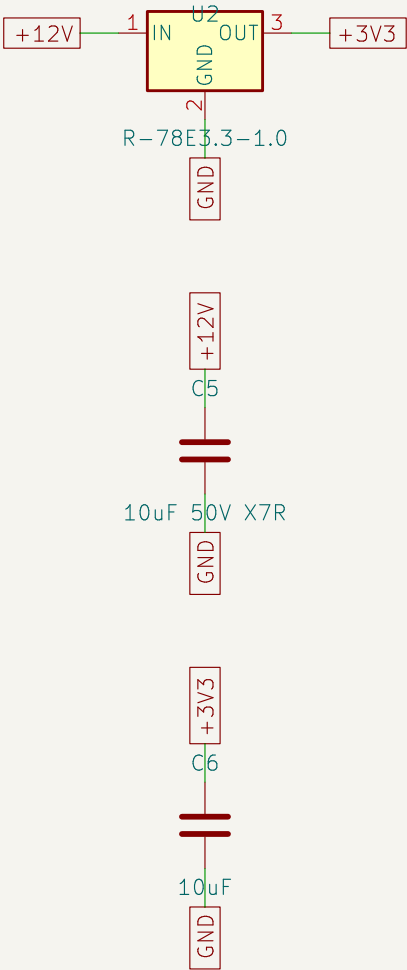
Date: 2026-09-13

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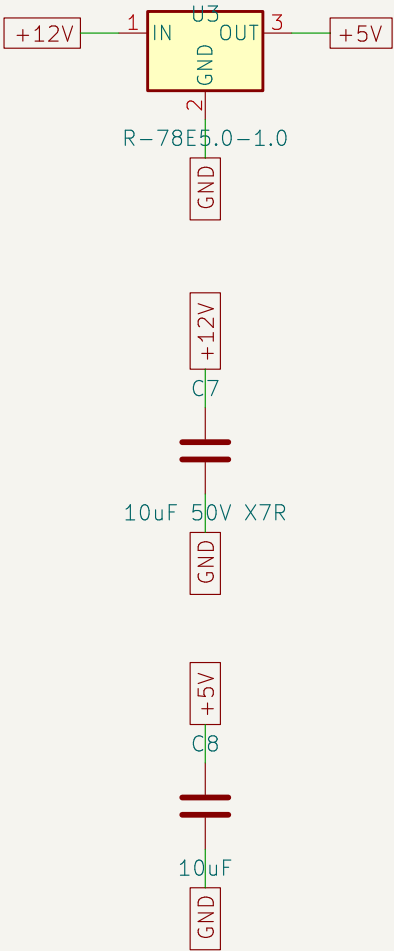
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3.3V REGULATOR



5V REGULATOR (SENSOR)



Palm-vein door lock – ESP32-S3, MOSFET lock driver, 12 V adapter

Sheet: /3.3V & 5V REGULATORS/  
File: power\_reg.kicad\_sch

**Title: palm-scan Rev 2 – 3.3V & 5V REGULATORS**

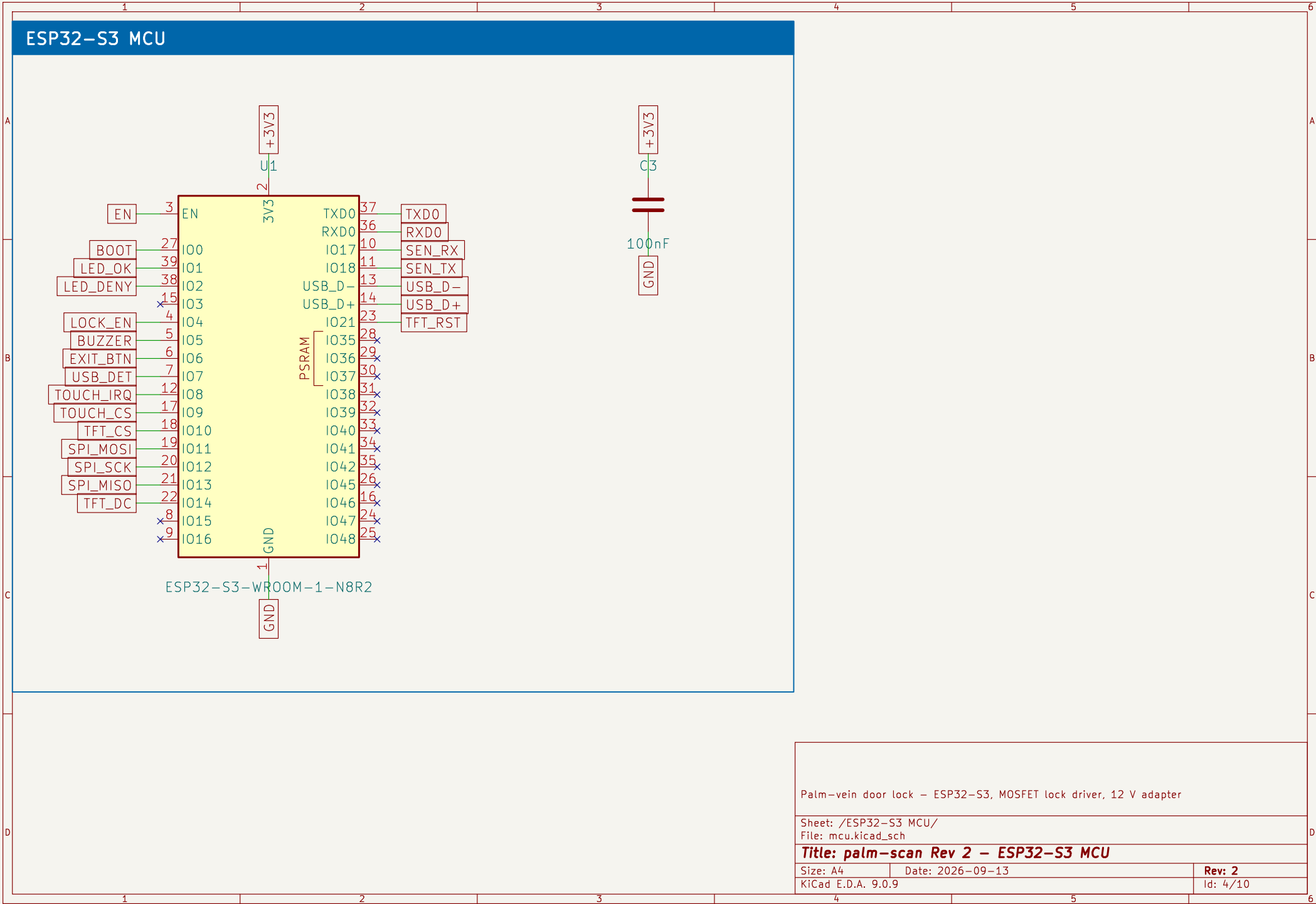
Size: A4

Date: 2026-09-13

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Palm-vein door lock – ESP32-S3, MOSFET lock driver, 12 V adapter

Sheet: /ESP32-S3 MCU/  
File: mcu.kicad\_sch

**Title: palm-scan Rev 2 – ESP32-S3 MCU**

Size: A4

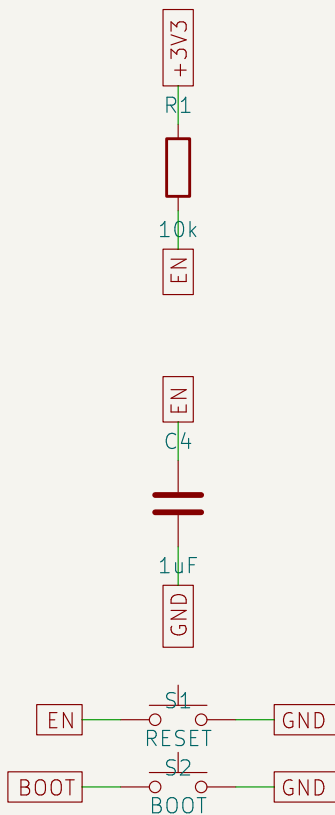
Date: 2026-09-13

Rev: 2

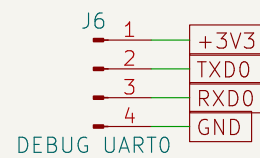
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## RESET & BOOT



## DEBUG UART



Palm-vein door lock – ESP32-S3, MOSFET lock driver, 12 V adapter

Sheet: /RESET, BOOT & DEBUG UART/  
File: reset\_boot.kicad\_sch

**Title: palm-scan Rev 2 – RESET, BOOT & DEBUG UART**

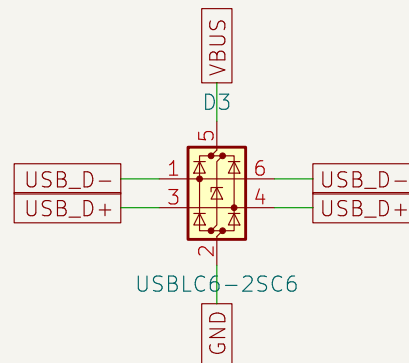
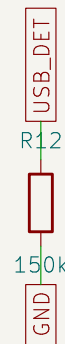
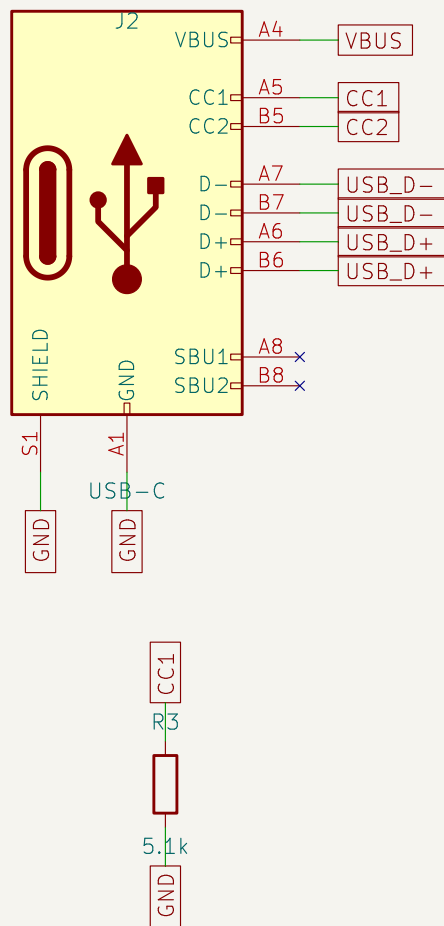
Size: A4 Date: 2026-09-13

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# USB-C (PROGRAMMING)



Palm-vein door lock - ESP32-S3, MOSFET lock driver, 12 V adapter

Sheet: /USB-C/  
File: usb.kicad\_sch

**Title: palm-scan Rev 2 - USB-C**

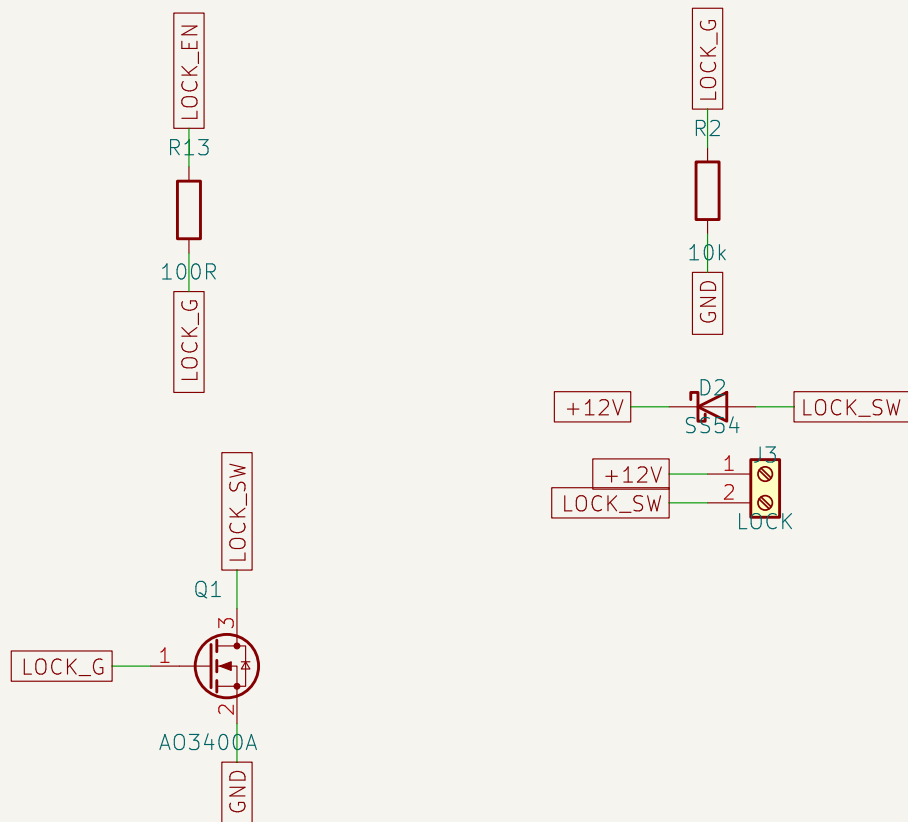
Size: A4 Date: 2026-09-13

KiCad E.D.A. 9.0.9

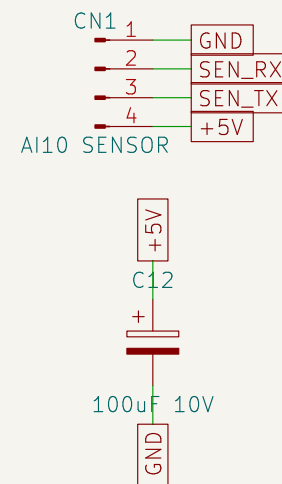
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## LOCK DRIVER



## PALM SENSOR



Palm-vein door lock - ESP32-S3, MOSFET lock driver, 12 V adapter

Sheet: /LOCK DRIVER & SENSOR/

File: lock.kicad\_sch

**Title: palm-scan Rev 2 - LOCK DRIVER & SENSOR**

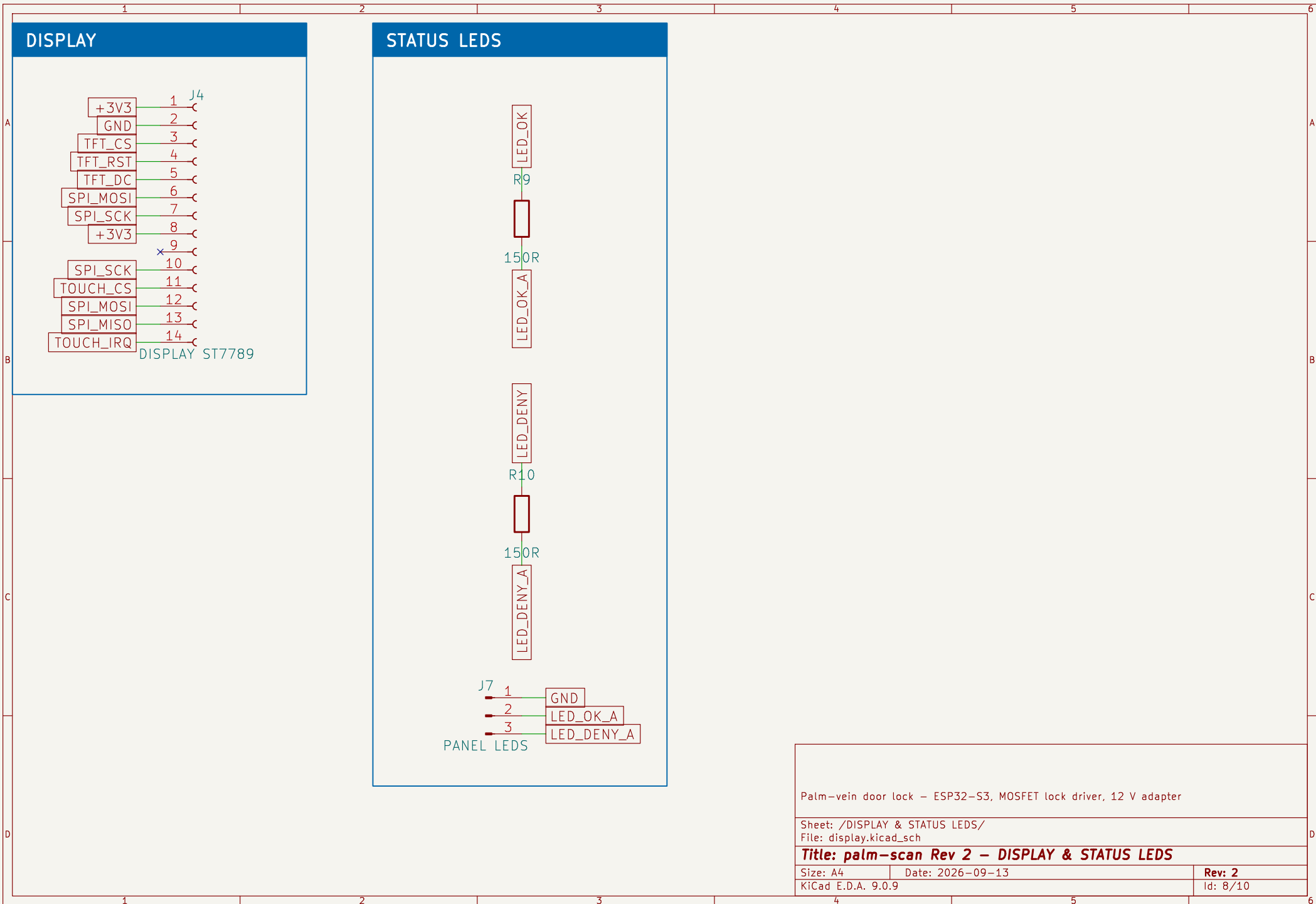
Size: A4

Date: 2026-09-13

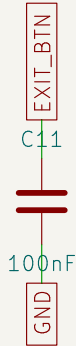
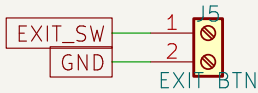
Rev: 2

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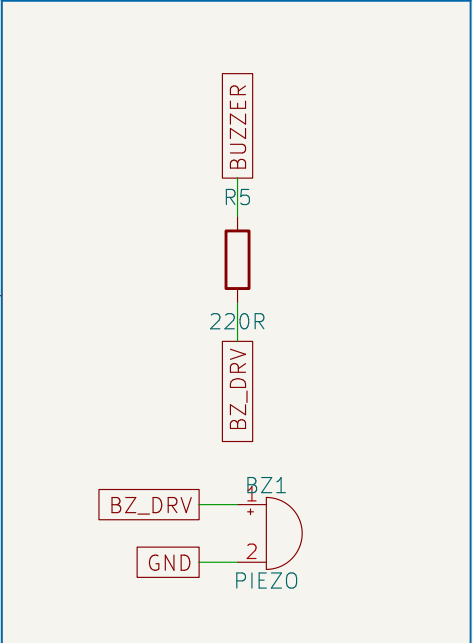
EXIT BUTTON



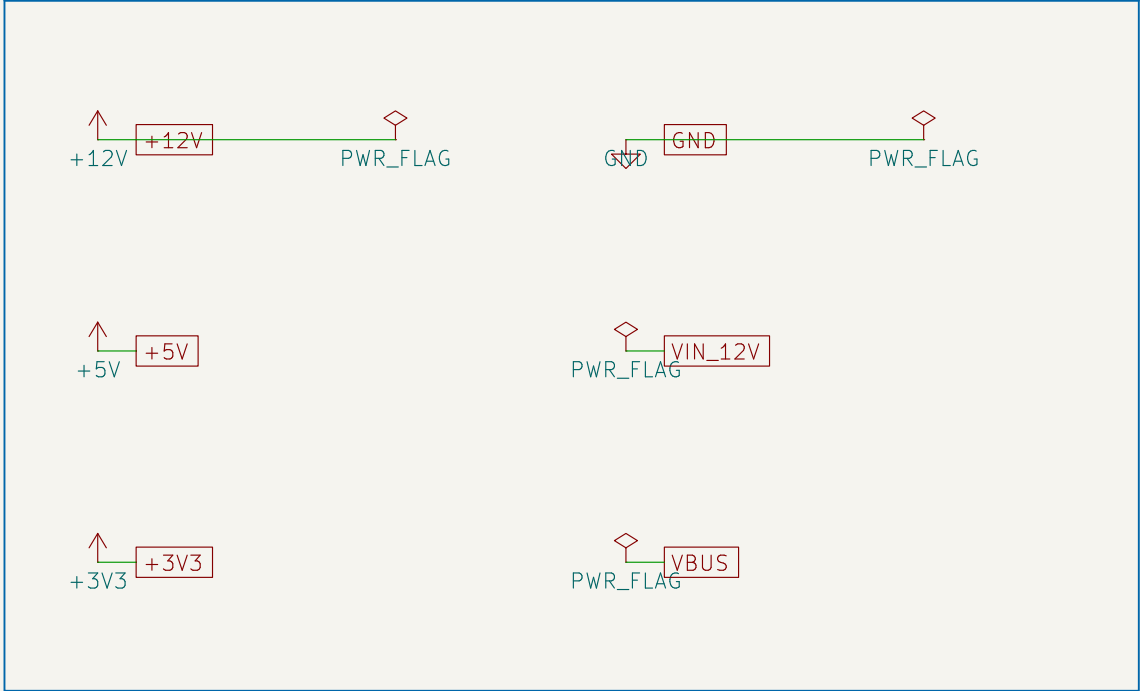
Palm-vein door lock – ESP32-S3, MOSFET lock driver, 12 V adapter

|                                              |                  |          |
|----------------------------------------------|------------------|----------|
| Sheet: /EXIT BUTTON/<br>File: exit.kicad_sch |                  |          |
| Title: palm-scan Rev 2 – EXIT BUTTON         |                  |          |
| Size: A4                                     | Date: 2026-09-13 | Rev: 2   |
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BUZZER



POWER FLAGS



Palm-vein door lock – ESP32–S3, MOSFET lock driver, 12 V adapter

Sheet: /BUZZER & POWER FLAGS/  
File: buzzer.kicad\_sch

**Title: palm-scan Rev 2 – BUZZER & POWER FLAGS**

|                    |                  |           |
|--------------------|------------------|-----------|
| Size: A4           | Date: 2026-09-13 | Rev: 2    |
| KiCad E.D.A. 9.0.9 |                  | Id: 10/10 |